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Tunable Q-factor wavelet transform of acoustic emission signals and its application on leak location in pipelines

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ABSTRACT

The tunable Q-factor wavelet transform (TQWT) is a discrete wavelet transform realized by the iterative two-channel filter group and the discrete Fourier transform (DFT). In this paper, the TQWT is applied to the extraction of leakage signals of the water pipeline network. By using the maximum principle of kurtosis, the Q-factor and the number of decomposition layer were determined to overcome the disadvantages of prefixed Q-factor and number of decomposition layers. The combination of kurtosis and envelope entropy was used to select the subband with abundant information. And then, correlation analysis was conducted further to locate the leakage point. Furthermore, a laboratory water pipe and an outdoor actual water pipe system were built to study the leakage signal. By analyzing the leakage signal from both experimental system, it can be found that the leakage signal characteristics of different leakage points for actual water pipe are all in the first layer, and the positioning error of each leakage point is within 1%. It indicates that TQWT can effectively extract the leakage components from the leakage signal of low SNR and has a higher location accuracy of the leak point.

1. Introduction

Pipeline is one of the most significant and economical ways for water, gas and oil transportation, and it plays a great role in the chemical and petrochemical industries, and urban water and gas supply system [1,2]. However, the leakage of pipeline cannot avoid happening due to some inevitable factors, such as erosion, aging, welding defects and third-party interference. Pipeline leakage may result in economic losses, environment contamination and even casualties. Therefore, it is of great significance to conduct timely and effective leakage detection.

In recent years, many kinds of approaches have been proposed to the leakage detection and location [3–7]. Among these methods, the acoustic emission technique has become one of the most prevalent method due to its real-time, in-service, relatively low cost, easy implementation. The acoustic emission signal carries the information about leak source, such as location and size. These information can be obtained by processing the acoustic emission signals.

A number of effective algorithms have been proposed to identify and locate the pipeline leakage, such as cross-correlation method, high-order cumulant method, least mean square (LMS) adaptive method, etc [3,4,7,8]. Cross-correlation method is the most widely used method for pipeline leak location due to its advantages of simplicity and accuracy in leak location. However, the effectiveness of cross-correlation method is reduced by the existence of noise, high energy interferers, and signal dispersion.

Actually the acoustic emission signals induced by the leakage of pipeline are non-stationary. The mutually independent and stationary assumption make the cross-correlation method cannot be effectively applicable to the non-stationary characteristic of acoustic emission signal. Therefore appropriate non-stationary signal processing methods must be employed to extract the time-delay estimation of non-stationary acoustic emission signals.

Since both time and frequency domain signal processing of AE can provide valuable information, a simultaneous time–frequency analysis proves to be quite effective. In recent years, many time-frequencies processing techniques has been widely studied for their effective feature extraction in non-stationary signals, such as short-time Fourier transform, Wigner-Ville distribution, empirical mode decomposition (EMD), and wavelet transform (WT) [9–12]. Among these methods, WT becomes one of the most outstanding techniques for its multi-resolution analysis capability, and has been widely used for acoustic source location [13–17]. Mostafapour developed a leak location algorithm with one AE sensor [16,17]. The acoustic leakage signals were decomposed into high and low frequencies by wavelet transform, and the peak magnitudes of wavelet transform were used to determine the arrival

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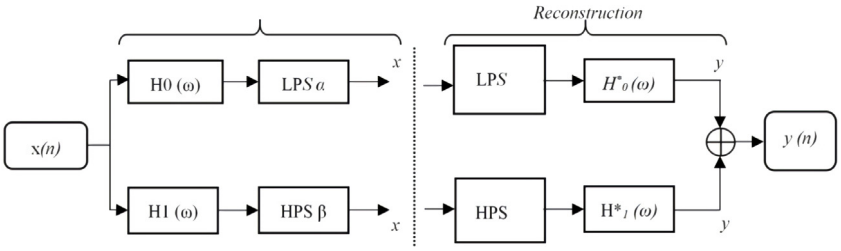


Fig. 1. Block diagram of analysis and synthesis filter banks for the TQWT.

times of flexural and extensional modes. The algorithm were validated by leakage location experiments in gas-filled pipe under environment with high-level of noises. Davoodi et al. presented a leak location algorithm combining of wavelet transform, filtering and cross-correlation analysis [16]. Mostafapour et al. used the theories of wavelet transform and cross-time frequency spectrum to locate AE source with frequency-varying wave velocity in plate. Yang et al. proposed an acoustic source location method in plate based on multiple signal classification method and wavelet transform [18]. They estimated the direction of arrival waves using the multiple signal classification and the time delay of the arrival wave was gained using the continuous wavelet transform. Jiao et al. proposed a source location method with one AE sensor [15]. The acoustic signal was processed by Gabor wavelet transform, and the arrival times of the different modes were determined by peaks of the magnitude of the wavelet transform. The above research demonstrated that the acoustic source location can be improved to a certain extent by the wavelet analysis of acoustic emission signals.

It has been known that wavelet transform is a type of inner product transform that analyzes non-stationary features in the signal through a predetermined wavelet basis. The similarity between the wavelet basis and the non-stationary features in the signals plays an important role in its effectiveness. Therefore, an appropriate wavelet is necessary for the diagnosing and feature extraction in non-stationary signal. For example, Ahadi investigated the effect of mother wavelet on the performance of WT, it was shown that by properly chosen mother wavelets, the AE events can be detected and identified precisely, and leakage AE signals in water-filled plastic pipes can also be effectively recognized.

However, the effectiveness of traditional WT is weakened by a few inevitable limitations. Firstly, the traditional WT is sensitive to translation of the input signal, which causes waveform distortion in extracting impulse features. Secondly, the traditional wavelet transform is commonly known as constant-Q transform, whose Q-factor cannot be adjusted automatically and flexibly. An alternative technique to overcome these limitations is the tunable Q-factor wavelet transform (TQWT). TQWT is a recently developed wavelet transform designed, and the Q-factor can be easily and continuously adjusted. The motivation for the TQWT is the efficient representation of signals that exhibit some degree of oscillatory behavior. It is thought that, by adjusting the Q-factor of the wavelet transform so as to match the oscillatory behavior of the signal under analysis, a more efficient signal representation can be obtained [19–23].

Compared with traditional wavelet transforms, the TQWT can offer excellent property in tuning the Q-factor to match the signal oscillatory behavior. Thus, the TQWT is presented as an effective tool that can be adapted to the analysis of leakage AE signals. In this paper, Tunable Q-factor wavelet transform is used to analyze the leakage AE signals. Firstly, the effect of Q-factor and decomposition level on the decomposition of leakage AE signals are studied, which based on features of kurtosis. Secondly, a method of extracting instantaneous leak response is proposed, which can be used to locate leak by conducting cross correlation analysis. Thirdly, some experiments are carried out to validate the proposed method.

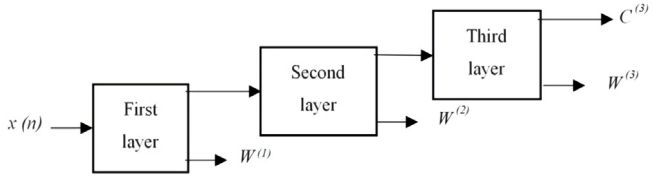


Fig. 2. The filter group of TQWT.

2. Principle of TQWT

The TQWT is basically a fully-discrete wavelet transform. It facilitates analysis of oscillatory signals with easily adjustable parameters. The main input parameters of TQWT consist of Q-factor Q, total oversampling rate r and number of levels of decomposition j. The parameter Q controls the number of oscillations of the wavelet. Theoretically, Q-factor is defined as the ratio of the center frequency of the wavelet with its bandwidth.

$$Q = f_0/BW \tag{1}$$

where, f_0 is the center frequency of the wavelet, BW is the bandwidth.

The parameter r controls the undesired excessive ringing in order to localize the wavelet in time without affecting its shape. The filters of TQWT are nonrational transfer functions which can be easily specified in frequency-domain for its efficient implementation.

The concept of the TQWT is simple. It is proposed using perfect reconstruction over-sampled filter banks with real valued scaling factor. The associated analysis and synthesis filter banks for the TQWT are shown in Fig. 1 and Fig. 2, respectively (where LPS and HPS represent low and high frequency pass filters, respectively, H∗ is the conjugate complex). The tunable-Q wavelet transform (TQWT) is implemented by iteratively applying the two-channel filter bank on its lowpass channel. The Fig. 1 shows a layer of decomposition and synthesis process of the finite long discrete time signals. $H_0(\omega)$ and $H_1(\omega)$ respectively represent the low pass filter and the high pass filter, which constitute the complete reconstruction filter of double channel filter group. The scale of high pass β and low pass α satisfy the Eqs. (2) and (3) respectively.

$$\beta = 2/(Q+1) \tag{2}$$

$$\alpha = 1 - \beta/r \tag{3}$$

As shown in Fig. 2, through repeatedly applying filter bank shown in Fig. 1 to low frequency channel, it can realize the adjustable Q-factor wavelet transform of finite long signal. The parameters $N^{(j)}$ represents the length of the input signal of j layer input signal. $N_0^{(j)}$ and $N_1^{(j)}$ represent the length of the subband signal produced by j layer filter group. Because the high pass subband constitutes the wavelet coefficient, the length of wavelet subband is $N_1^{(j)}$, the low pass subband produced by j layer filter is $c^{(j)}$. Setting $N_0^{(j)} \approx \alpha^j N$, $N_1^{(j)} \approx \alpha^{j-1}\beta N$, the following formula can be obtained.

$$N_0^{(j)} = 2round(\frac{\alpha^j}{2}N), \quad N_1^{(j)} = 2round(\frac{\beta\alpha^{j-1}}{2}N) \tag{4}$$

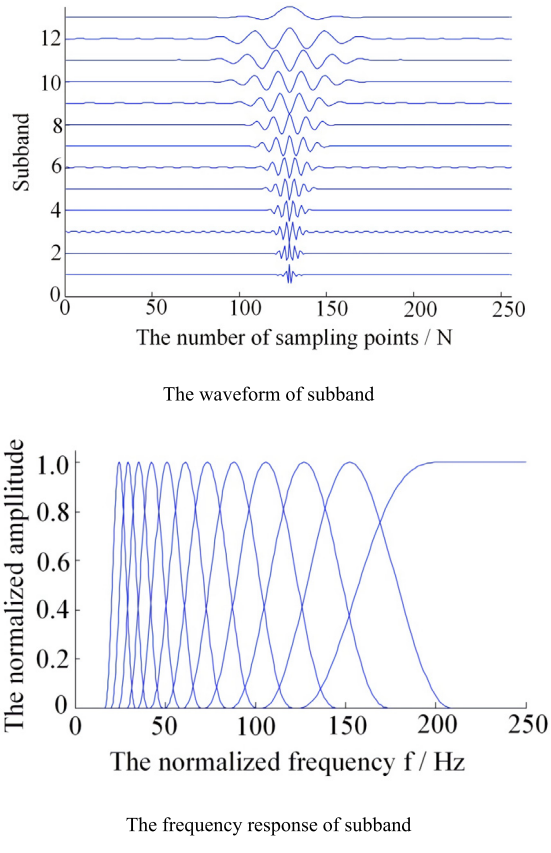


Fig. 3. The waveform and frequency response of each wavelet subband (Q = 3, R = 3, J = 12).

If $1 \le j \le J$, then,

$$N^{(1)} = N, N^{(j)} = N_0^{(j-1)}, 2 \le j \le J \tag{5}$$

Setting χ as the input signal, which length is N, the wavelet transform of the adjustable Q-factor can be derived as,

$$C^{(0)} \leftarrow DFT\{x\}, \left\{C^{(j)}, W^{(j)}\right\} \leftarrow AFB\left(C^{(j-1)}, N_0^{(j)}, N_1^{(j)}\right)$$
$$w^{(j)} \leftarrow DFT^{-1}\left\{W^{(j)}\right\}, c^{(J)} \leftarrow DFT^{-1}\left\{C^{(J)}\right\} \tag{6}$$

where, $1 \le j \le J$, AFB represents the decomposition filter group, $c^{(j)}$ and $w^{(j)}$ are the low-pass and high-pass subband produced by the jth layer filter group.

Meanwhile, the inverse transformation can be derived as,

$$C^{(J)} \leftarrow DFT\left\{c^{(J)}\right\}, W^{(j)} \leftarrow DFT\left\{w^{(j)}\right\}$$
$$C^{(j-1)} \leftarrow SFB\left(C^{(j)}, W^{(j)}, N^{(j)}\right), y \leftarrow DFT^{-1}\left\{C^{(0)}\right\} \tag{7}$$

where, $1 \le j \le J$, SFB represents construction filter group.

The essence of the TQWT is the Q maintain constant, and the quality factor can be preset. Setting Q = 3, J = 12, the wavelet time domain waveform is shown in Fig. 3(a). From the figure we can know, with the number of decomposition layers increase, the time of wavelet vibration becomes longer. The Fig. 3(b) shows the frequency response of subband. From the figure we can know, the frequency response represent some filter groups with non-constant bandwidth and the adjacent band are not orthogonal. With the number of decomposition layers increase, the center frequency and the bandwidth decrease. Both can be written as

$$f_c = \alpha^{(j)} \frac{2-\beta}{4\alpha} f_s, (j = 1, 2, \ldots, J) \tag{8}$$

$$B_w = \frac{1}{2}\beta\alpha^{(j-1)}\pi, (j = 1, 2, \ldots, J) \tag{9}$$

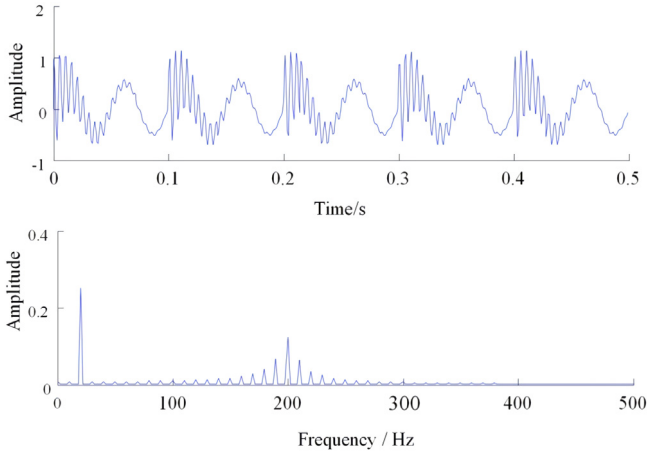


Fig. 4. The time domain and spectrum distribution of simulation signal.

If j is integer and f_s is the sampling frequency, the maximum of scale parameter J can be derived as

$$J \approx \frac{\log(\beta \cdot N/8)}{\log(1/\alpha)} \tag{10}$$

In order to validate the effectiveness of TQWT algorithm extracting oscillation signals, the following simulation signals are constructed.

$$x_1 = e^{-40t_1}\sin(400\pi t_2), \quad x_2 = \sin(40\pi t_2) \tag{11}$$

where, $t_1 = \mathrm{mod}((0:511)/fs, 1/10)$, $t_2 = (0:511)/fs$, sampling frequency $fs = 1024$ Hz. Synthetic signal satisfies the equation $x = x_1 + x_2$. Its time domain and spectrum distribution are shown in (Fig. 4).

The simulation signal is processed by wavelet multi-scale decomposition and adjustable q-factor wavelet transform respectively. The wavelet basis function is "db6" and the decomposition layer is 2. The parameters of TQWT are $Q = 1$, $J = 2$. The process result is shown in Figs. 5 and 6.

The Figs. 5 and 6 show that TQWT is a good way to separate the two different frequency components of the signal, but the wavelet multi-scale decomposition will cause the frequency aliasing and be unable to separate two frequency components correctly.

3. Methodology

3.1. Leak localization method based on TQWT

The pipeline leaks localization method based on TQWT mainly includes the following steps. Firstly, the quality factor Q was determined based on the kurtosis maximum value after the preprocessing of each acoustic signal to be analyzed. Secondly, according to the maximum difference of kurtosis, the number of decomposition layers is determined. Thirdly, each acoustic signal to be analyzed is decomposition by TQWT according to the determined parameters. Fourth, A small wave layer of leakage characteristics was selected according to kurtosis and envelope entropy. The wavelet layer was conducted intercorrelation analysis to determine the location of the leak point. The location process is shown in Fig. 7.

3.2. Determination of TQWT parameter

The shape of wavelet times domain waveform and the quality factor of the frequency domain were determined by Q-factor. The scale J determines the extension of the wavelet time domain waveform and the frequency domain filter. The wavelet function corresponding to the small Q-factor has a smaller periodic number and a narrower bandwidth in the time domain. The wavelet function corresponding to

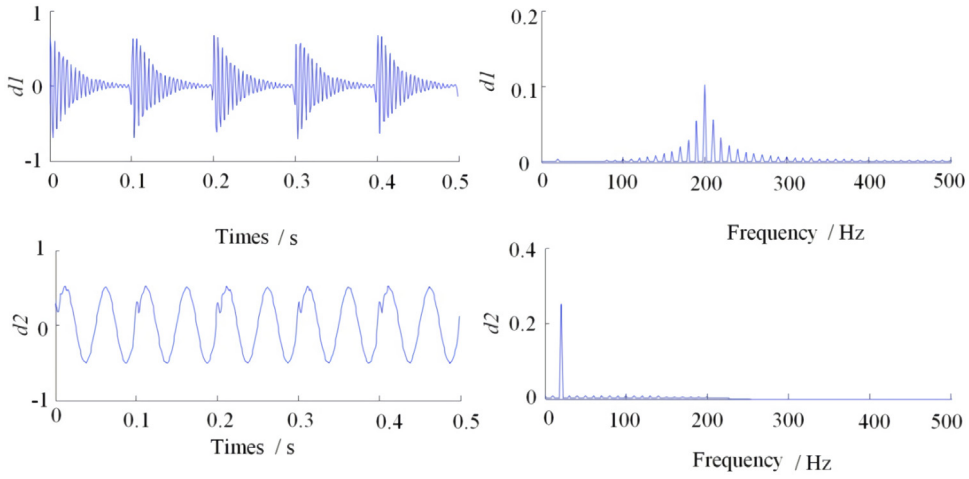


Fig. 5. The time domain and spectrum distribution of each layer after TQWT decomposition of the simulation signal.

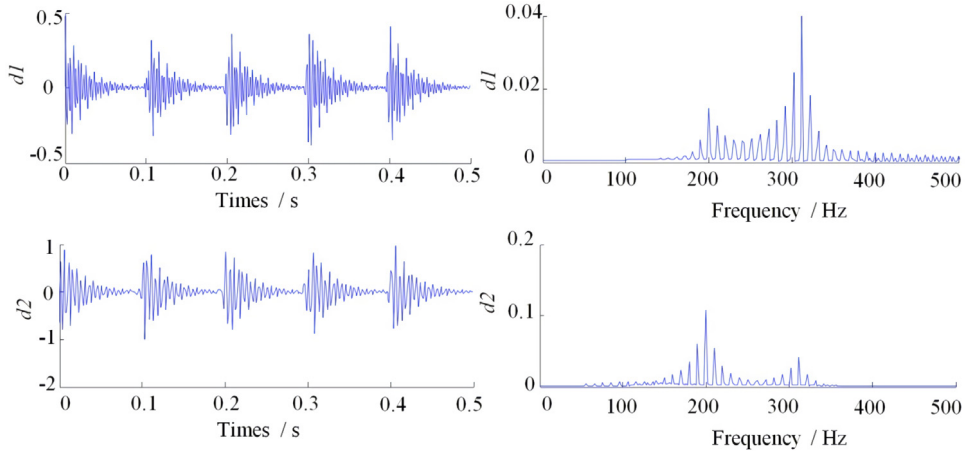


Fig. 6. The time domain and spectrum distribution of each layer after analyzed by wavelet multiscale decomposition.

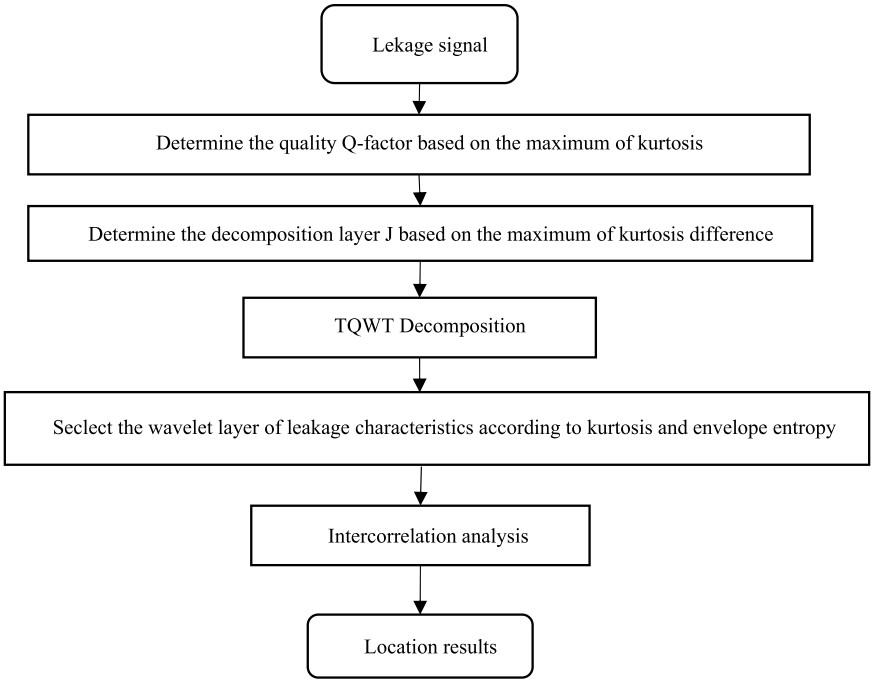
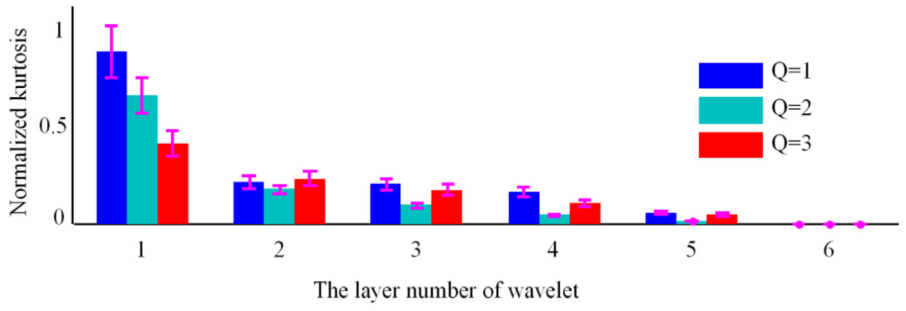
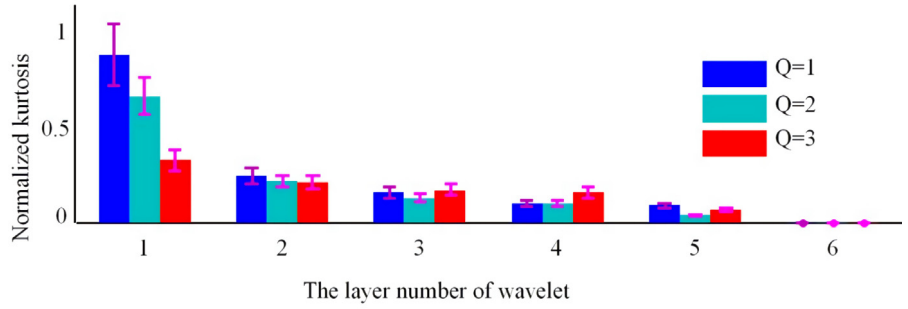


Fig. 7. The flow chart of pipeline leak location based on TQWT.



(a) The left channel



(b) The right channel

Fig. 8. Comparison of the normalized kurtosis of each wavelet layer under different Q factors.

the large Q-factor has a higher periodic number and a wider bandwidth in the time domain. Therefore, the impact composition and harmonic components can be extracted from small Q-factor and large Q-factor. Because the leakage signal is a non-stationary vibration signal generated by the flow impact tube wall, the impact components in the signal can be extracted with the small Q-factor. The optimal TQWT parameters are discussed by experimental method below.

3.2.1. Analysis of valve leakage signal

A simulated pipeline was built in the laboratory. The valve was used to simulate the leakage source of the pipeline. The sampling rate was 5000 Hz, and the sampling time was 5 s.

(1) The determination of optimal quality factor Q

The pipeline leakage generate a non-stationary random signal. There are many differences between different moment of leakage signal. In order to accurately extract the leakage components of the signal, the signal should be preprocessed before TQWT processing to select the appropriate quality factor Q. At the same time, the leakage signal of the pipeline is an broadband and instantaneous shock signal and have a low quality factor. Therefore, this paper studies the ability of TQWT to decompose the leakage signals at 1, 2 and 3 respectively (see Fig. 8).

Kurtosis is the parameter describing the peak of the waveform. When the signal has multiplied impact compositions, the kurtosis value increases obviously.

When the pipeline is leaking, the turbulent jet formed by the leaking gas has an impact on leakage hole wall. The larger of the kurtosis amplitude indicate the impact components in the signal has a large proportion. Usually, effective leakage signals are often contained in signals that have more impact. After TQWT decomposition of the leakage signal, the higher the kurtosis of the wavelet layer means the more leakage is extracted. Consequently, when the quality factor Q is equal to 1, it is best to extract leakage signal after TQWT decomposition, according to the principle of kurtosis get to the maximum.

(2) The determination of decomposition layer number J

After determining the quality factor Q = 1, the maximal TQWT decomposition layer is 9 according to formula (9). The kurtosis variation of each layer were studied below in order to determine the optimal decomposition layer, when the decomposition layer ranges from 4 to 9.

The greater the difference between the two adjacent layers of wavelet, the stronger the ability of the wavelet to extract the leakage components. As shown in Fig. 9, the difference of kurtosis get to the maximum when the decomposition layer is 6. Consequently, the decomposition layer of TQWT can be determined as 6.

3.2.2. Analysis of hydrant valve leakage signal

In order to simulate the pipe leakage signal more realistic, the hydrant valve was used to simulate pipeline leakage source. The hydrophone is used to collect the leakage signal. The sampling rate is 5000 Hz, and the single sampling time is 5 s.

(1) The determination of optimal quality factor Q

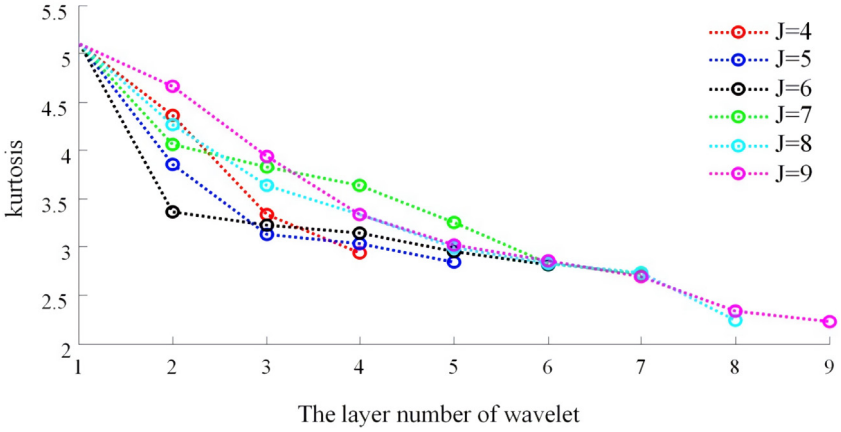
After TQWT decomposition of the leakage signal, the greater of the wavelet kurtosis indicates that more leaked ingredients are taken out. According to the principle of maximal kurtosis, it has the best effect to extract leakage signals after TQWT decomposition when quality factor Q was designated 1 as shown in Fig. 11 (see Fig. 10).

(2) The determination of decomposition layer number J

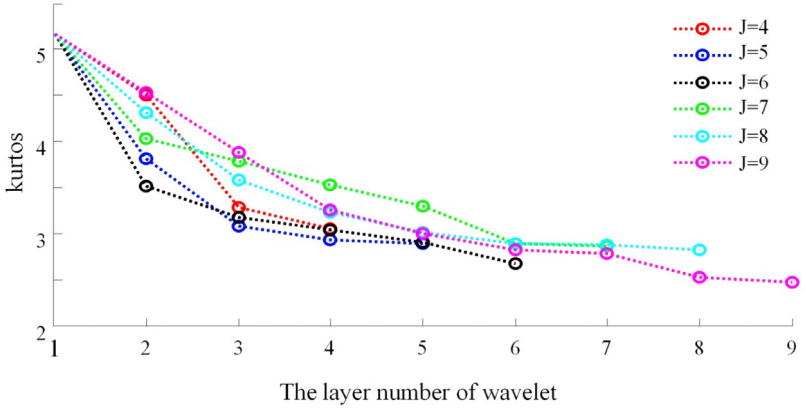
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The greater the difference between the two adjacent layers of wavelet, the stronger the ability of the wavelet to extract the leakage components. As shown in Fig. 13, the difference of kurtosis get to the maximum when the decomposition layer is 6. Consequently, the decomposition layer of TQWT can be determined as 6 (Fig. 12).

The above discussion indicates that the TQWT decomposition parameters of the leakage signals collected by the laboratory simulation pipeline are consistent with the leakage signals collected by the hydrant



（a）The left channel



(b) The right channel

Fig. 9. The contrast of the kurtosis of each wavelet layer in different decomposition layers.

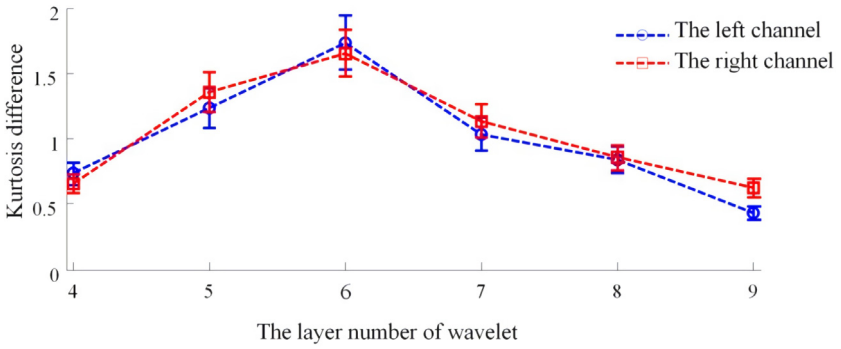


Fig. 10. The kurtosis difference of the first two layers of wavelets under different layers of decomposition.

pipeline. When Q = 1 and J = 6, the time domain waveform of its subband and frequency response curve are shown in Fig. 14.

4. Application in leak location

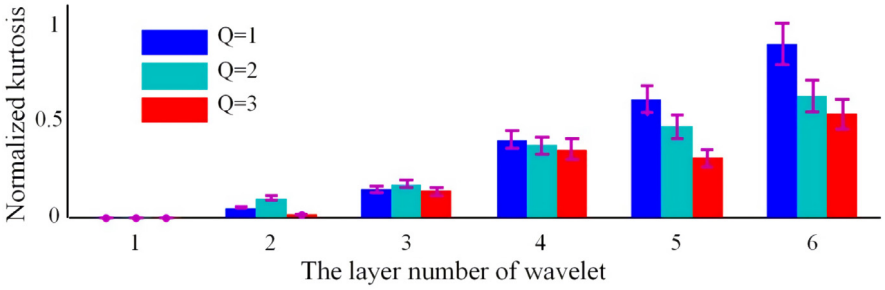
In order to verify the effectiveness of the proposed method, the leakage signals of the simulated water supply pipeline in laboratory and the actual water supply pipeline in the field are collected respectively.

4.1. The analysis of leakage signal of laboratory water pipe

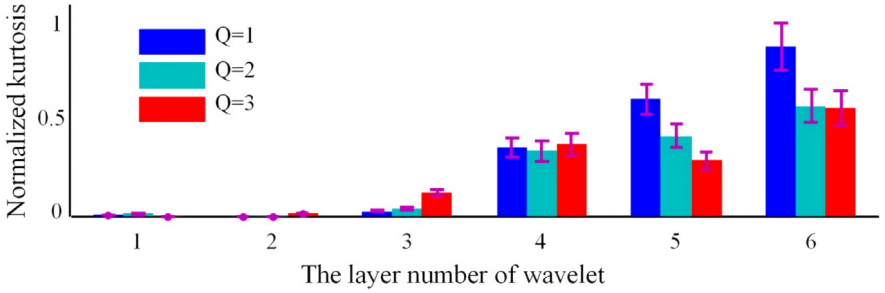
The length of the simulated water pipe is 18 m and there are three leaks, as shown in Fig. 15. One leak was opened to simulate the leakage every time. One of the three leakage signals was analyzed below. The time domain and spectrum distribution are shown in Fig. 16.

As shown in Fig. 15, the waveform of the leakage signal is extremely complex and has no periodic component. The distribution frequency band of the leakage signal is wider and concentrate on the range of 0–2000 Hz. TQWT decomposition results of the leakage signal are shown in the following figures (Fig. 17).

From Fig. 17, It can be seen that different wavelet layer signals reflect different characteristics of the original signal. In order to further extract the small wave layer with abundant leakage characteristics, the kurtosis and effective value entropy of each layer are analyzed. The leakage signal of pipeline is characterized by uncertainty, complexity and non-linearity. Entropy is a characteristic index of uncertainty and

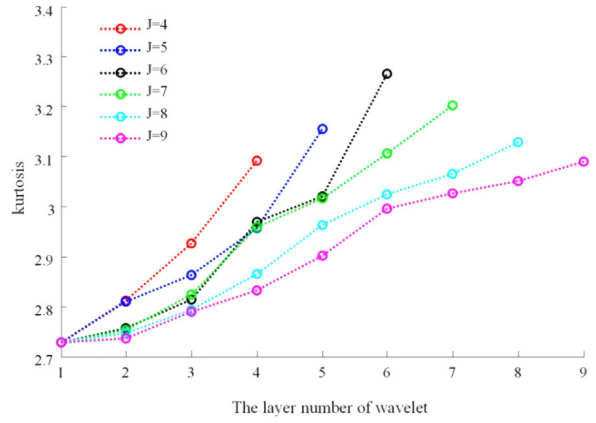


(a) The left channel

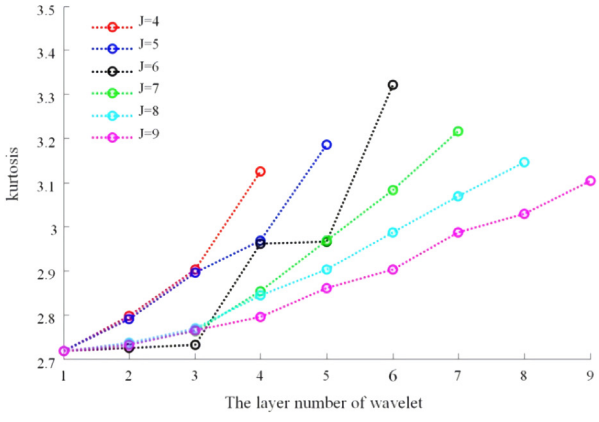


(b) The right channel

Fig. 11. Comparison of the normalized kurtosis of each wavelet layer under different Q.



(a) The left channel



(b) The right channnel

Fig. 12. The contrast of the kurtosis of each wavelet layer in different decomposition layers.

Table 1
The location results of different leak point.

Leak point number	1		2		3	
The wavelet layer of leakage signal	1	1	1	1	1	1
The correlation coefficient	0.161	0.238	0.299	0.386	0.242	0.238
Time delay/s	0.0038	0.0036	0.0012	0.0012	−0.0022	−0.0022
Leakage location/m	6.475	6.595	8.035	8.035	10.075	10.075
Error/m	−0.115	0.005	−0.055	−0.055	−0.035	−0.035

signal complexity of signal status. The entropy of envelope spectrum is proposed based on the definition of information entropy, which can describe the mutation of leakage signal. Therefore, the abundance of the leakage components in the signal can be represented by the envelope entropy representation signal.

By analyzing the kurtosis of each wavelet layer and the curve of the envelope spectrum entropy curve can be found, the kurtosis and envelope spectrum entropy of the first layer wavelets are the largest. Therefore, it can infer that the first layer of wavelet signal contains information of abundant leakage characteristics. By conducting inter-correlation analysis for the first layer of wavelet, the results are shown in Fig. 18.

The leakage signal is decomposed by TQWT and the leakage characteristic frequency band was extracted by kurtosis and envelope entropy as shown in Fig. 19. The correlation coefficient peak obtained by correlation analysis is more obvious. Furthermore, the estimated time of delay is closer to the theoretical value. All above illustrate that the leakage signal with low SNR is decomposed by TQWT, which can extract the leakage feature effectively. Furthermore, the leakage characteristic frequency band can be accurately extracted by kurtosis and envelope entropy. In order to further verify the applicability of this method, the signal of three different leakage points in the laboratory is analyzed, and the results are shown in the Table 1.

It can be found from the Table 1 that the leakage signal characteristics of different leakage points for the laboratory water pipe are all in the first layer, and the positioning error of each leakage point is within 1%.

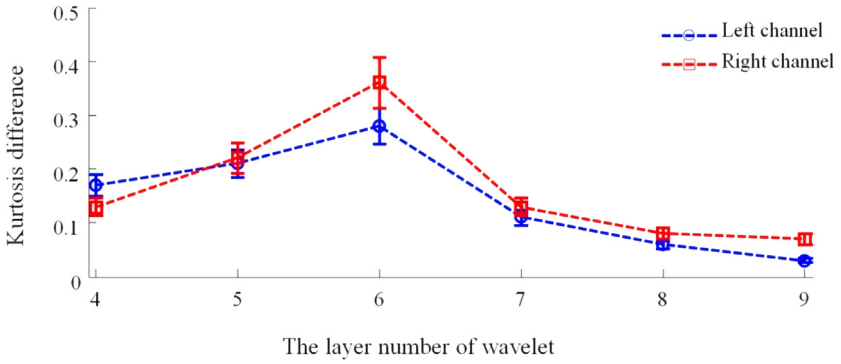
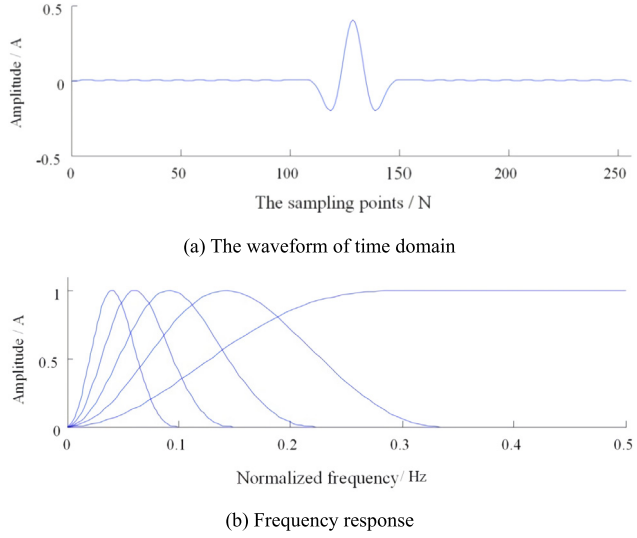


Fig. 13. The kurtosis difference of the first two layers of wavelets under different layers of decomposition.



(a) The waveform of time domain

(b) Frequency response

Fig. 14. The time domain waveform and frequency response of the first subband of TQWT wavelet (Q = 1, R = 3, J = 5).

4.2. The analysis of leakage signal of actual water pipe

The length of the actual water pipe is 100 m and there are three leaks. Similarly, one leak was opened to simulate the leakage every time. One of the three leakage signals was analyzed below. The time domain and spectrum distribution are shown in Fig. 20.

As shown in the Fig. 20, the waveform of the leakage signal is extremely complex and has no periodic component. The distribution frequency band of the leakage signal is wider and concentrate on the range of 0–1500 Hz. TQWT decomposition results of the leakage signal are shown in Fig. 21.

By analyzing the kurtosis of each wavelet layer and the curve of the envelope spectrum entropy curve can be found, the kurtosis and envelope spectrum entropy of the first layer wavelets are the largest. Therefore, it can infer that the first layer of wavelet signal contains information of abundant leakage characteristics. By conducting inter-correlation analysis for the first layer of wavelet, the results are shown in Fig. 22.

The leakage signal is decomposed by TQWT and the leakage characteristic frequency band was extracted by kurtosis and envelope entropy as shown in Fig. 23. The correlation coefficient peak obtained by correlation analysis is more obvious. Furthermore, the estimated time of delay is closer to the theoretical value. All above illustrate that the leakage signal with low SNR is decomposed by TQWT, which can extract the leakage feature effectively. Furthermore, the leakage characteristic frequency band can be accurately extracted by kurtosis and envelope entropy. In order to further verify the applicability of this method, the signal of three different leakage points in the laboratory is analyzed, and the results are shown in the Table 2.

Table 2
The location results of different leakage point.

Leak point number	1		2		3	
The wavelet layer of leakage signal	5	6	6	6	6	6
The correlation coefficient	0.305	0.332	0.124	0.215	0.233	0.117
Time delay/s	0.04426	0.04417	0.02358	0.02349	0.0048	0.00612
Leakage location/m	21.78	21.84	34.96	35.024	46.93	46.07
Error/m	0.48	0.54	−0.04	0.024	0.33	0.07

It can be found from the Table 2 that the leakage signal characteristics of different leakage points for actual water pipe are all in the first layer, and the positioning error of each leakage point is within 1%.

5. Conclusion

In this paper, the influence of Q factor and J number of decomposition layer on leakage components in extracted signals is studied. The laboratory plumbing test system was set up and the reliability of the laboratory water pipe is verified by comparing the real water pipe test system data. The characteristic parameters such as kurtosis and envelope entropy used to characterize the leakage signal were studied and selected as the basis for the selection of the wavelet layers. By analyzing the leakage signal results from both experimental system, it indicates that the characteristic parameters such as kurtosis and envelope entropy can be used to characterize the leakage signal. The TQWT is applied to the extraction of low SNR leakage signals. By decomposing the leakage signals into multiple wavelet layers, the separation of signal and noise is realized. Similarly, by analyzing the leakage signal results from both experimental system, TQWT can effectively extract the leakage components from the leakage signal of low SNR and has a higher location accuracy of the leak point. It can be found that the leakage signal characteristics of different leakage points for actual water pipe are all in the first layer, and the positioning error of each leakage point is within 1%.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

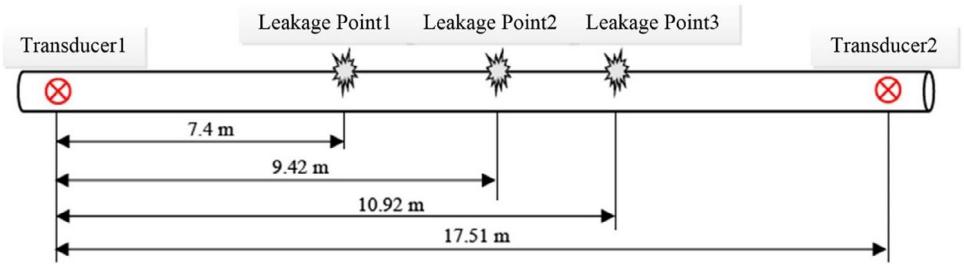


Fig. 15. The laboratory water pipe to analyze leakage signal.

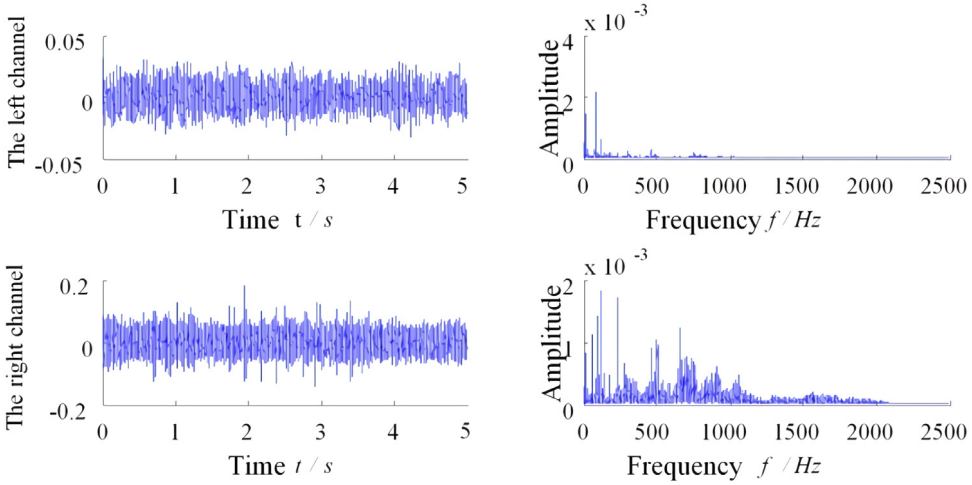


Fig. 16. Left and right channel time domain and spectrum distribution of leakage signal.

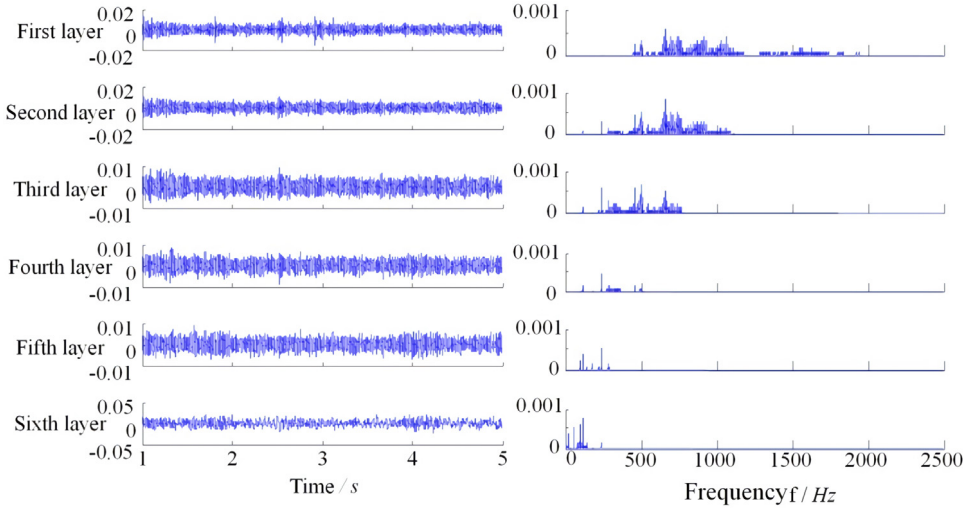
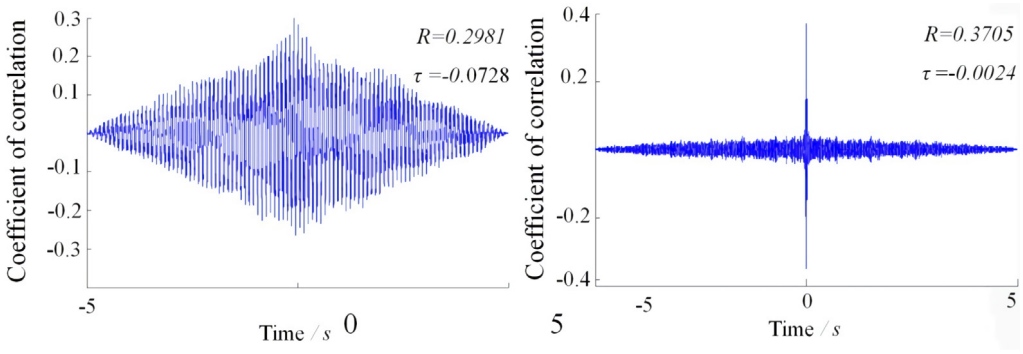


Fig. 17. The time domain and spectrum distribution of each wavelet layer after the leakage signal TQWT decomposition.

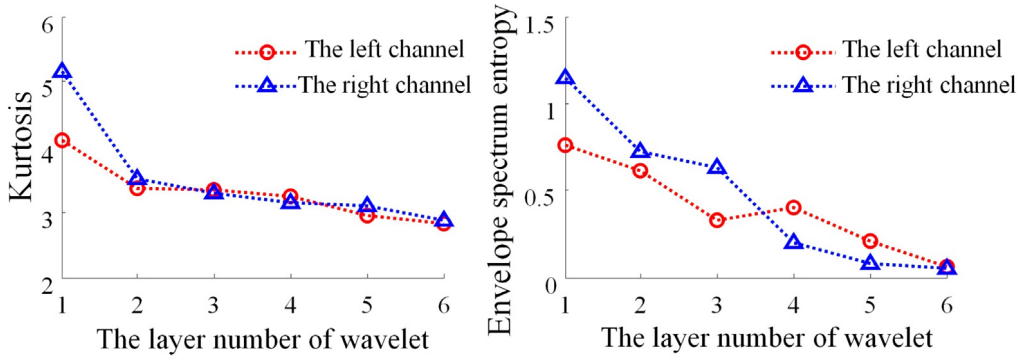
CRediT authorship contribution statement

Meng Tao: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing. **Wang Deguo:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing. **Jiao Jingpin:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing. **Li**



(a) The original signal (b) The wavelet signal of first layer

Fig. 18. The correlation analysis results of leakage signal before and after decomposition.



(a) kurtosis (b) Envelope spectrum entropy

Fig. 19. The curve of each wavelet layer kurtosis and the entropy change of the envelope after decomposition of the leakage signal.

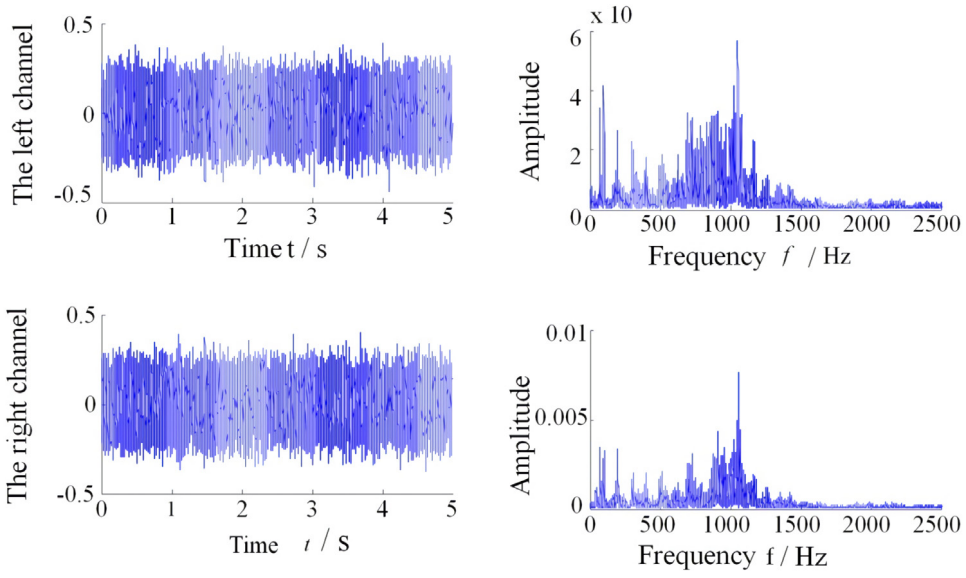


Fig. 20. Left and right channel time domain and spectrum distribution of leakage signal.

Xiaolong: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing.

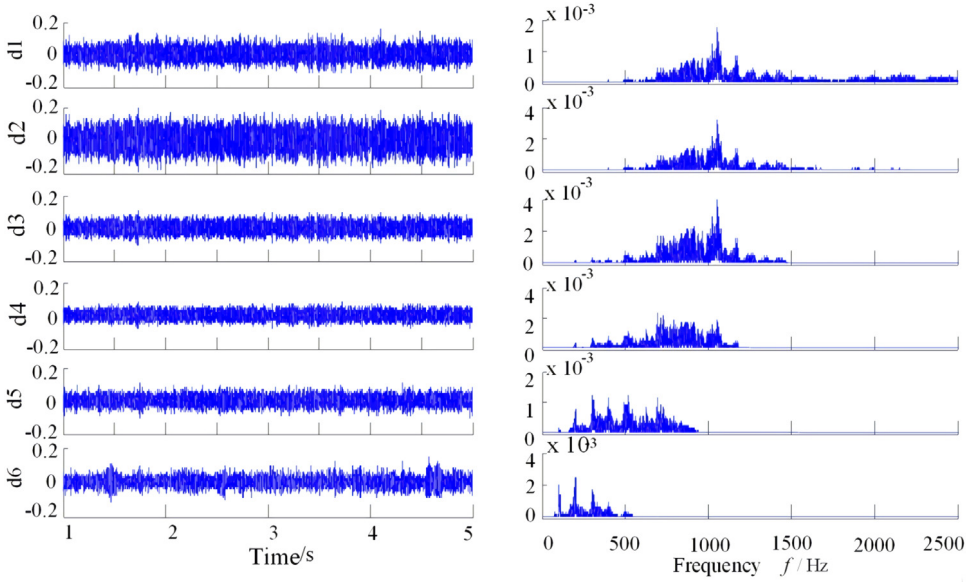
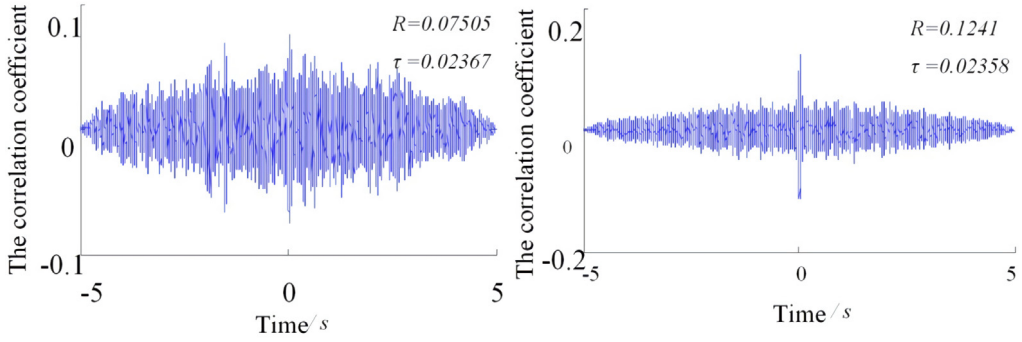
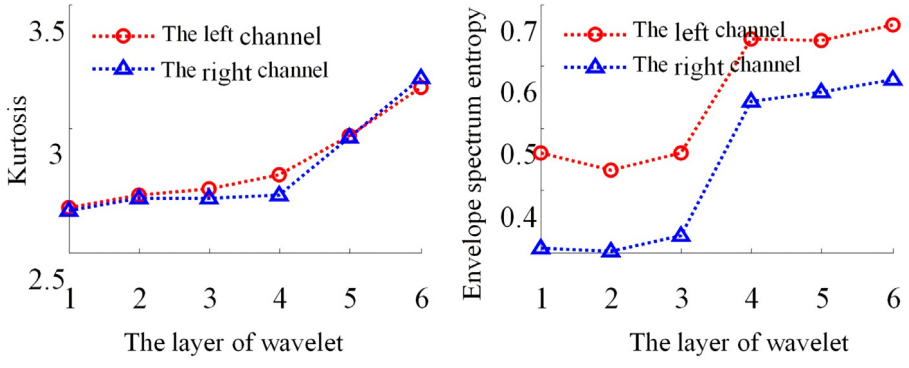


Fig. 21. The time domain and spectrum distribution of each wavelet layer after the leakage.



(a) The original signal (b) The first layer of the wavelet signal

Fig. 22. The correlation analysis results of leakage signal before and after decomposition.



(a) Kurtosis (b) Envelope spectrum entropy

Fig. 23. The curve of each wavelet layer kurtosis and the entropy change of the envelope after decomposition of the leakage signal.

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Ethical approval and consent to participate

Approved.

Consent for publication

Approved.

Availability of supporting data

We can provide the data.

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References

[1] Y. Guo, T. Meng, D. Wang, H. Tan, R. He, Experimental research on the corrosion of X series pipeline steels under alternating current interference, Eng. Fail. Anal. 78 (2017) 87–98.
[2] Y. Guo, T. Meng, D. Wang, T. Meng, S. Liu, R. He, Comprehensive risk evaluation of long-distance oil and gas transportation pipelines using a fuzzy petri net model, J. Nat. Gas Sci. Eng. 33 (2016) 18–29.
[3] Z. Poulakis, D. Valougeorgis, C. Papadimitriou, Leakage detection in water pipe networks usinga Bayesian probabilistic framework, Probab. Eng. Mech. 18 (2003) 315–327.
[4] J. Sun, Q. Xiao, J. Wen, Natural gas leak location with K–L divergence-based adaptive selection of Ensemble Local Mean Decomposition components and high-order ambiguity function, J. Sound Vib. 347 (2015) 232–245.
[5] P.S. Murvay, I. Silea, A survey on gas leak detection and localization techniques, J. Loss Prev. Process Ind. 25 (2012) 966–973.
[6] Y.F. Xu, J.M. Liu, W.D. Zhu, Accurate and efficient calculation of discrete correlation functions and power spectra, J. Sound Vib. 347 (2015) 246–265.
[7] S.Y. Li, Y.M. Wen, P. Li, Leak location in gas pipelines using cross-time–frequency spectrum of leakage-induced acoustic vibra, J. Sound Vib. 333 (2014) 3889–3903.
[8] J. Sun, Q. Xiao, J. Wen, Natural gas pipeline leak aperture identification and location based on local mean decomposition analysis, Measurement 79 (2016) 147–157.
[9] M.S. Kim, S.K. Lee, Detection of leak acoustic signal in buried gas pipe based on the time–frequency analysis, J. Loss Prev. Process Ind. 22 (2009) 990–994.
[10] D. Ozevin, J. Harding, Novel leak localization in pressurized pipeline networks using acoustic emission and geometric connectivity, Int. J. Press. Vessels Pip. 92 (2012) 63–69.
[11] C. Guo, Y. Wen, P. Li, Adaptive noise cancellation based on EMD in water-supply pipeline leak detection, Measurement 79 (2016) 188–197.
[12] J. Sun, Q. Xiao, J. Wen, Natural gas pipeline small leakage feature extraction and recognition based on LMD envelope spectrum entropy and SVM, Measurement 55 (2014) 434–443.
[13] Z. Yang, Z. Xiong, M. Shao, A new method of leak location for the natural gas pipeline based on wavelet analysis, Energy 35 (2010) 3814–3820.
[14] A. Mostafapour, S. Davoodi, M. Ghareaghaji, Acoustic emission source location in plates using wavelet analysis and cross time frequency spectrum, Ultrasonics 54 (2014) 2055–2062.
[15] J. Jiao, B. Wu, C. He, Application of wavelet transform on modal acoustic emission source location in thin plates with one sensor, Int. J. Press. Vessels Pip. 81 (2004) 427–431.
[16] A. Mostafapour, S. Davoodi, Continuous leakage location in noisy environment using modal and wavelet analysis with one AE sensor, Ultrasonics 62 (2015) 305–311.
[17] S. Davoodi, A. Mostafapour, Gas leak locating in steel pipe using wavelet transform and cross-correlation method, Int. J. Adv. Manuf. Technol. 70 (2014) 1125–1135.
[18] H. Yang, T.J. Shin, S.k. Lee, Source location in plates based on the multiple sensors array method and wavelet analysis, J. Mech. Sci. Technol. 28 (1) (2014) 1–8.
[19] I.W. Selesnick, Wavelet transform with tunable Q-factor, IEEE Trans. Signal Process. 59 (8) (2011) 3560–3575.
[20] J. Luo, D. Yu, A kurtosis-guided adaptive demodulation technique for bearing fault detection based on tunable-Q wavelet transform, Meas. Sci. Technol. 24 (2013) 1–11.
[21] G. Cai, X. Chen, Z. He, Sparsity-enabled signal decomposition using tunable Q-factor wavelet transform for fault feature extraction of gearbox, Mech. Syst. Signal Process. 41 (2013) 34–53.
[22] S. Patidar, R.B. Pachori, Classification of cardiac sound signals using constrained tunable-Q wavelet transform, Expert Syst. Appl. 41 (2014) 7161–7170.
[23] S. Patidar, R.B. Pachori, Automatic diagnosis of septal defects based on tunable-Q wavelet transform of cardiac sound signals, Expert Syst. Appl. 42 (2015) 3315–3326.